Control of Camber Morphing Wing

Project report submitted in partial fulfillment of the Requirements for the Award

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BACHELOR OF TECHNOLOGY

In

MECHANICAL ENGINEERING

By

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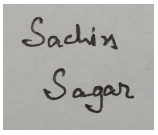
(2023-24)

DECLARATION

I certify that this dissertation's ideas, designs, experimental work, results, analyses, and conclusions are entirely my effort, except where otherwise indicated and acknowledged. I certify that the work is original and has not been previously submitted for assessment in any other course or institution except where specifically stated.



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CERTIFICATE

This is to certify that the project report titled "Control of camber morphing wing," submitted by Sachin Sagar (ME19B159) to the Indian Institute of Technology, Madras, for the award of the degree of Bachelor of Technology in Mechanical Engineering, is a bonafide record of the project work carried out by him under my supervision. The contents of this report, in whole or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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Without their patience and support, I would not have been able to come as far as I have come in my academic life.

ABSTRACT

This project aims to model the temporal dynamics and design a control system for a “Double Corrugated Variable Camber (DCVC)” airfoil. This is part of a larger project funded by DRDO to design a camber morphing wing for a UAV and benchmark its performance against conventional flap-based designs under wind tunnel testing.

The trailing edge of the airfoil needs to be morphed using a hingeless mechanism. For the morphing to work seamlessly and hold the desired angle under environmental loading, active actuation is needed. This is achieved using a servo motor and will also be tested during wind tunnel testing. The controller is needed to hold the morphing section in a particular orientation under varying wind loads and at various angles of attack.

The project is pursued under two verticals: modeling the morphing section and designing a controller. A modal approach is followed for the former. Mode shapes of the morphing section are obtained using Ansys Modal analysis. A linearized system model is built on Matlab by considering the structural and aerodynamic interactions of the wing under various wind loads and angles of attack.

A model predictive controller (MPC) is conceived that determines the optimal control move given the physical constraints of the structure. To measure the morphing section in real time, a video-based image processing approach is employed. The centerline of the morphing section is tracked using a color-based clustering algorithm. A Kalman Filter is used to estimate the shape of the aerofoil camber line.

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NOMENCLATURE

T	Total Kinetic Energy	ρ_f	Fluid density
U	Total Potential Energy	α	Pitch angle
q_j	Generalized Displacement of jth DoF	U, U_∞	Free Stream Velocity
Q_j	Generalized Force of jth DoF	l	Length of the morphing section
w	Beam deflection	β	Flap angle
Φ	Mode shape/ Basis function	τ	Servo time constant
ξ	Generalized displacement	G	Servo Gain
w' or θ	Slope	T_i	i_th Theodeorsens constant
ξ_i	ith deflection state	b	Chord length/2
E	Elastic Modulus	k	Time step
I	Moment of Inerta		

CHAPTER 1 - Introduction

1.1 Background and Motivation

The lift requirements during takeoff, climb, cruise, descent, and taxiing of an aircraft are different. The current norm is to use airfoil control surfaces like flaps, slats, and spoilers to alter the lift generated as per requirement. However, the discontinuity in the flow introduced due to a sudden change in the airfoil geometry causes flow separation at lower angles of attack, leading to inefficiencies and increased fuel consumption.

Active camber morphing wings provide an amicable solution to the above problem, as the airfoil geometry is altered without introducing any discontinuities. Morphing aircraft concepts are showing the potential to significantly reduce aircraft mass and fuel consumption and improve flight performance. With the abundance of Li-ion cells and the advances in composite materials, there is an increasing interest in developing morphing wings across the aerospace industry.

Morphing can be implemented either on the leading or the trailing edge. Trailing edge morphing can be achieved using:

- Four bar mechanisms
- Fish Bone Active Camber (FishBAC)
- Single and Double Corrugated Structures

The rigid section is usually made from aluminum, and the skin is made using elastomeric matrix composite (EMC). Actuation can be achieved using motors or smart actuators like macro fiber composites (MFCs).

1.2 Objectives and Scope of the Project

The use of compliant structures implies that the wing's behavior under aerodynamic loads is modified. This paper aims to develop a dynamic model for the morphing wing under variable wind loads and angles of attack. Following are the objectives for this project

a) **Modelling**: A physics-based model is developed for the system. The equations of motion are developed using the Hamiltons principle, with aerodynamic loads acting as the forcing functions. Non-linearities are ignored, and orthogonal mode shapes are used to model the deformation. The aerodynamic loads are estimated using linear plate theory under quasi-steady state conditions.

b) **Control Design:** The model is checked for stability, and a controller is designed to actuate the trailing edge to the desired morphing angle under wind tunnel testing conditions. Trailing edge morphing is measured in real time using a camera and fed to the controller. A model predictive controller is used to determine optimal input, and a Kalman Filter is used for full-state feedback.

The work in this project was carried out under the assumption that the model and the controller would be used to control the wing in a wind tunnel. The model reduction and linearization do not hold under in-flight conditions.

CHAPTER 2 - Literature Review and Experimental Setup

2.1 Structural Model

Transverse vibrations of a loaded beam are modeled using the Euler-Bernoulli equation. For a homogenous beam (constant elastic modulus (*E*) and area moment of inertia (*I*)), an analytical expression can be obtained for the deflection as a function of position and time (*w(x,t)*). The mode shapes (*Φ(x)*) are mutually orthogonal, leading to independent equations of motion for the corresponding generalized displacements (*ξ(t)*).

The equations of motion can be obtained using Lagrange-d'Alembert equations:

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} + \frac{\partial U}{\partial q_j} = Q_j \tag{2.1}$$

For a complicated structure like the DCVC, the mode shape and corresponding natural frequencies can be found using finite element method on software like Ansys Modal Analysis.

2.2 Aerodynamic model

The flow is expected to separate near the trailing edge. An unsteady pressure distribution composed of a circulatory and a non-circulatory component is defined. This pressure distribution is integrated over the camber line of the morphing section to find Q_j,

$$L_{morph} = \int_0^l \Delta P(x,t) \cos\theta \, dx \tag{2.2}$$

where $\Delta P(x,t)$ is the pressure difference between the lower and upper surfaces of the airfoil, θ is the slope of the morphing section.

It is common practice to ignore the time-dependent circulatory component for simplicity. This leads to a quasi-steady state model, such that the aerodynamic forces depend on time only through deflection.

$$P_{nc}(x,t) = -2\rho_f \frac{\partial^2 w}{\partial t^2}\sqrt{x(l-x)} + \frac{\rho_f U_\infty (2x-l)}{\sqrt{x(l-x)}}\left(\frac{\partial w}{\partial t} + U_\infty \frac{\partial w}{\partial x}\right) \qquad (2.3)$$

2.3 Actuator and Flap Angle Dynamics

A servo motor is essentially a DC motor with a large gear ratio. It forms a second-order system with respect to the angular position.

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{1}{\tau} \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ \frac{G}{\tau} \end{bmatrix} u(t) \qquad (2.4)$$

where x(t) is a vector of β_shaft and its derivative, and u(t) is voltage input.

A servo cannot reach the specified commanded angle under open loop conditions. At least a measurement feedback from a potentiometer or an incremental encoder is needed.

It is a common practice to model the servo motor with a position controller as a separate entity and consider it capable of achieving any commanded position.

The motor provides the required hinge moment to counter the self-weight and aerodynamic loads acting on the flap. However, the commanded flap angle is different from the achieved flap angle. For a rigid flap, a small oscillations model can be used to show that the flap angle, under aerodynamic loads, follows 2nd order dynamics given by:

$$C_\beta \dot{\beta} + I_\beta \ddot{\beta} - K_\beta (\beta_c - \beta) = M_\beta \qquad (2.5)$$

Here, I_β, C_β and K_β, are structural inertia, damping and stiffness of the flap and β_c is the commanded flap angle.

Theodorsen *et al.* (1949) [8] has found unsteady M_β using linear plate theory. Quasi steady-state aerodynamic moment is found by setting *C(k) = 1*.

$$M_\beta = -\frac{\beta\,\rho_f\,(T_5 - T_{10}\,(T_4 - T_{12}))\,U^2\,b^2}{\pi} + \frac{T_{11}\,\dot{\beta}\,\rho_f\,(T_4 - T_{12})\,U\,b^3}{2\,\pi} + \frac{T_3\,\ddot{\beta}\,\rho_f\,b^4}{\pi} - T_{12}\,\alpha\,\rho_f\,b^2 \tag{2.6}$$

Where T_i is the ith Theodorsens constant, these are functions of the airfoil geometry.

The expression is known to hold well for symmetric airfoils. Assuming piece-wise constant U and α. We observe a bias term in the commanded flap angle given by:

$$\beta_{\text{eq}} = \frac{T_{12}\,\alpha\,b^2\,\rho_f}{K_\beta} \tag{2.7}$$

This suggests a baseline moment is needed to hold the flap at a given angle for a given angle of attack. Bias is accommodated into the control system by appropriately shifting the setpoint.

2.4 Model Predictive Control and Estimation

Model Predictive Control (MPC) is a control input design methodology that simulates the system to predict its state, Np timesteps (prediction horizon). It would calculate the optimal control move if the setpoint were achieved in Nc timesteps (control horizon) from the current instant. The MPC solves an optimization problem at every instant. Physical constraints can be included as bounds on control input, its time derivative, and the setpoint. An MPC is tuned by adjusting cost functions associated with control input cost and setpoint error and changing Np and Nc.

State estimation is needed when many states need to be estimated, given a limited number of measurements. A Kalman Filter is such an estimator that combines the knowledge of the state with measurements while minimizing the error covariance matrix. A Kalman filter requires the observability and controllability matrices to be full rank.

2.5 Physical setup

Under a project grant from DRDO, the Vibration Control and Energy Harvesting Lab of IIT Madras has developed a morphing wing. A double corrugated variable camber structure is used to achieve trailing edge morphing of a NACA 641612 airfoil.

The corrugated structure is constructed from 0.5mm molded aluminum sheets and constitutes 30% of the chord length. The skin is made from an elastomeric matrix composite. The rigid section of the wing is made from 5mm machined aluminum and houses the power supply, servo motor, and transmission.

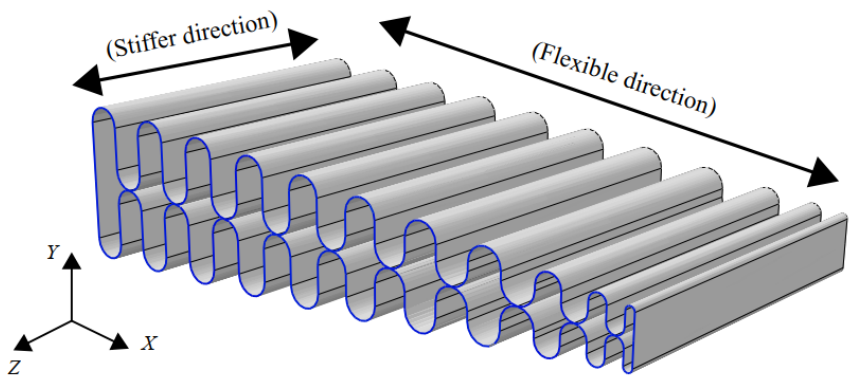

Figure 2.1: Double Corrugated structure with anisotropy [3]

The DCVC structure is stiff in the spanwise direction and compliant in the chordwise direction. Deepak et al. (2021) [3] have developed a large deflection model to predict the morphed shape in the flexible direction for a given actuation force.

Ongoing efforts include improving the cycle life of the DCVC by restricting the range of actuation and redesigning the bevel gear transmission. The team intends to test the morphing wing setup in a wind tunnel this fall.

2.6 Actuation

The wing is actuated by pulling on tendons attached to the free end of the last DCVC fold. These tendons are connected to a barrel, with a ring gear on one end. The pinion gear is connected to the servo motor. The RMCS-2016, a 255 kg-cm servo, is used with the RMCS-2301 motor driver and PID controller for actuation. An 18V DC power supply powers the setup. An incremental rotary encoder with a resolution of 9412 CPR is used for feedback.

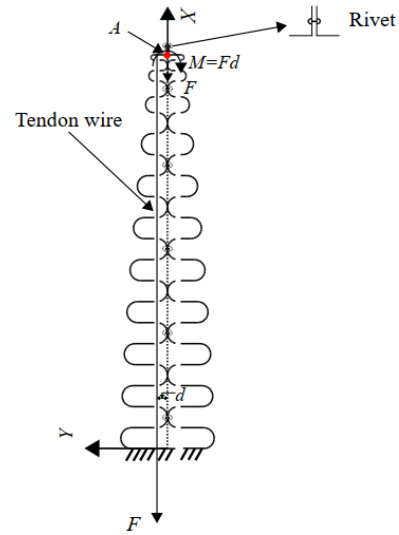

Figure 2.2: Actuation using tendons [3]

2.7 Implications for the Project

A typical airfoil with a flap comprises 3 degrees of freedom, viz—pitch, heave, and flap angle.

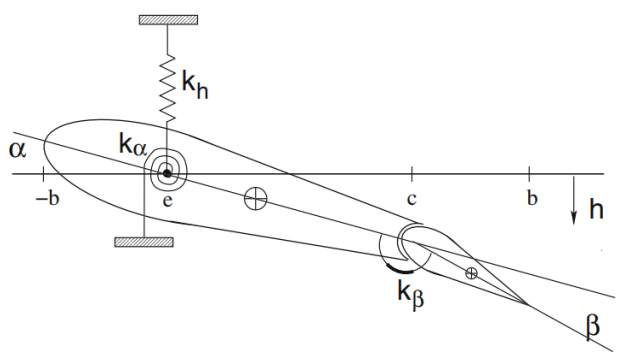

Figure 2.3: Schematic of a symmetric airfoil with a flap [4]

However, the pitch (α) and heave (h) degrees of freedom are artificially restrained for wind tunnel testing. The test will sweep across various wind speeds, angles of attack (pitch), and morphing angles. Hence:

- As U is piece-wise constant, the system is linear
- As pitch and plunge are restrained, a 1 DoF system is possible.

CHAPTER 3 - Modelling and Analysis

3.1 Modal analysis of the morphing section

To build a state space model of the continuously morphing section its mode shapes are used. These are obtained through finite element based numerical simulations on Ansys Modal Analysis.

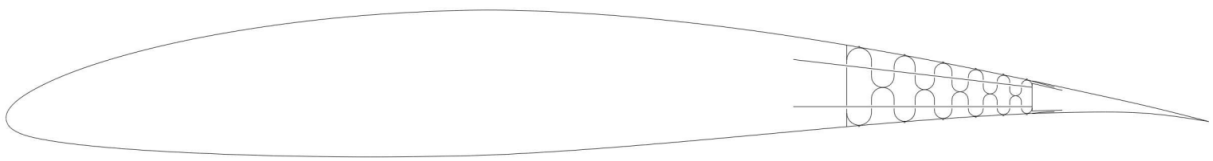

Figure 3.1: NACA 641612 airfoil with morphing trailing edge

NACA 641612 is an unsymmetric airfoil. The morphing section comprises 30% of the chord length. The geometry was simplified for modal analysis by truncating the rigid section and removing the airfoil skin and the tendons. 0.5mm thickness was given to the corrugated structure, and the material was specified as Al-6061. A multizone mesh was generated with an element size of 0.5 mm.

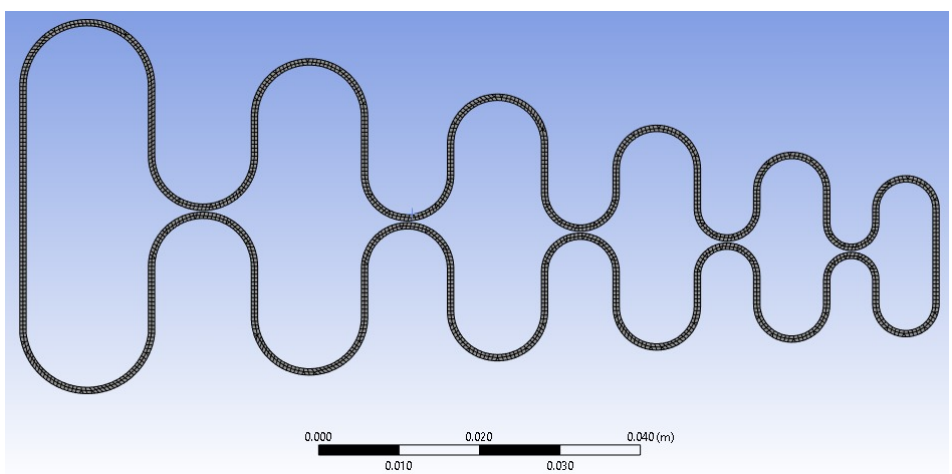

Figure 3.2: Morphing section, simplified and meshed with 2709 elements

3.2 Setup and Post-processing

Fixed support was specified on one end, and bounded contacts are specified at the rivet locations. The system is simulated for the first ten modes, and the directional deformation is calculated.

For control simulation, 1D mode shapes are needed. The shape of the morphing section is assumed to be represented by the centerline. Coordinates of the centerline deformation for transverse (bending) modes and the corresponding natural frequencies are extracted. A 'makima' spline fit is used for interpolation. It is desirable to use cantilever basis functions for the state space model. Hence a regression fit is done to express the DCVC mode shapes in terms of cantilever beam modes, and a transformation matrix is formed.

3.3 Results

a) Of the ten-mode shapes simulated, six were bending, and four were axial. Only transverse modes were deemed necessary for the current study.

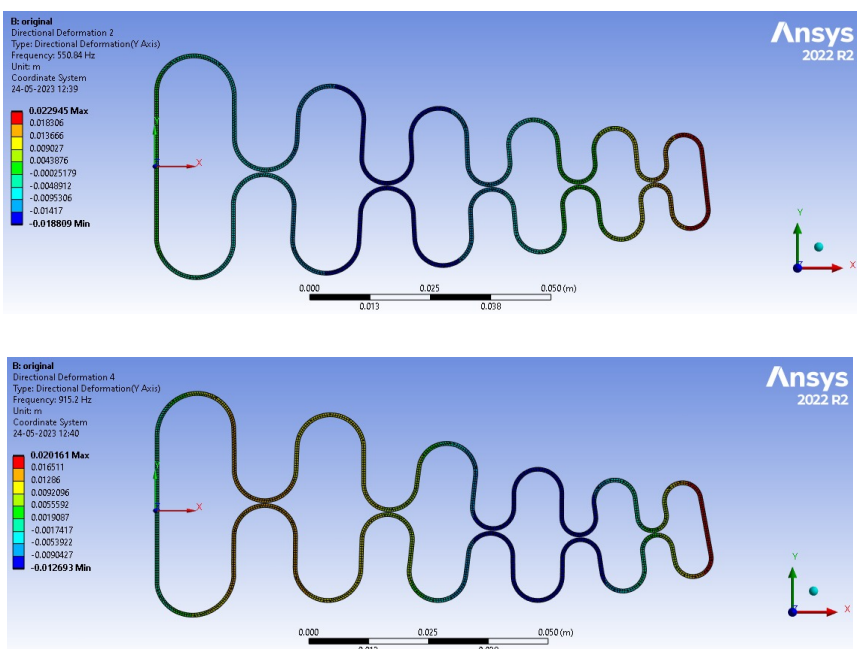

Figure 3.3: a) Mode 2 (550.84 Hz) and b) Mode 4 (915.2 Hz) for the DCVC structure

b) The regression fit revealed an increasing trend in normalized sum-square error for higher modes. However, performing an SVD of the transformation matrix revealed that the first three cantilever modes explained over 99% of the DCVC modes

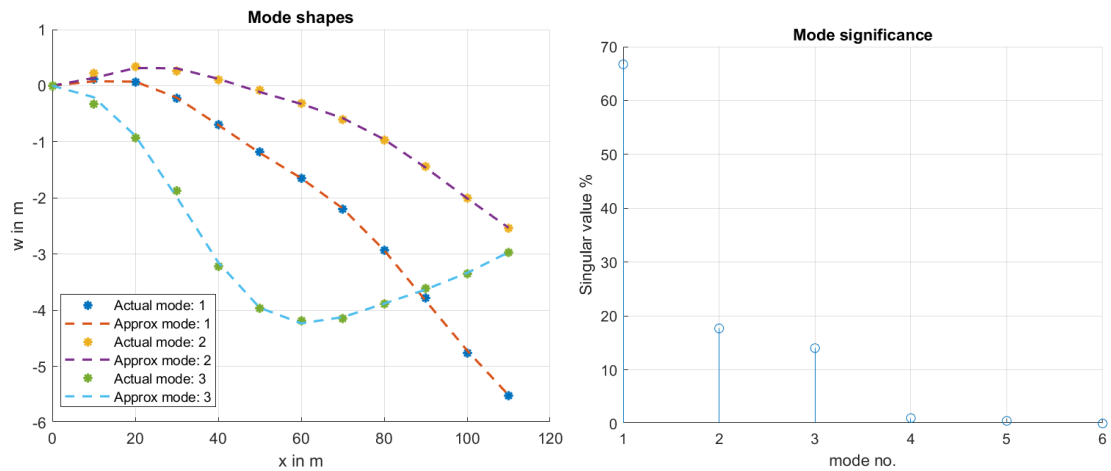

Figure 3.4: a) DCVC mode shapes approximated using cantilever beam modes.

b) Singular values of the transformation matrix

3.4 State Space formulation

Dynamics of the rigid flap are assumed to approximate the dynamics of the morphing section closely. The bias term is left out of the state space as it does not affect the system's dynamics and can be added during post-processing. The effect of morphing is introduced by augmenting the state space with deflection states corresponding to the first three modes of the state space.

$$X = \begin{pmatrix} \xi_1 & \xi_2 & \xi_3 & \dot{\xi}_1 & \dot{\xi}_2 & \dot{\xi}_3 & \beta & \dot{\beta} \end{pmatrix} \tag{3.1}$$

The augmented state space is found by appropriately stacking the deflection and flap angle states. Interaction between the two appears in the form of the input. Zhang and Shaw et al.

(2021) [7] have shown that the loading of the morphing section can be approximated by neglecting the time-dependent terms in the non-circulatory pressure distribution.

$$L_{morph} = \int_0^l \frac{\rho_f U_\infty^2 (2x - l)}{\sqrt{x(l-x)}} \frac{\partial w}{\partial x} dx = \frac{\rho_f U_\infty^2}{EI} \int_0^l \frac{(2x - l)}{\sqrt{x(l-x)}} x \, dx M_{in} \tag{3.2}$$

Further, $M_{in} = K_\beta . \beta_c$. However, as the aerodynamic loads have already been accounted for in the flap model, the forcing terms appear only in the deflection states in the B matrix.

The state matrices can be found in the appendix.

3.5 Morphing angle and flap angle

The morphing angle is a function of position. To relate it with the flap angle, we arbitrarily define β as $\arctan(y_5/x_5)$, where (y_5, x_5) is the position of the free ends center end of the morphing section.

3.6 Simulation Results

a. Modal deflection states

- All poles and zeros lie on the left half plane, indicating that the system is stable.

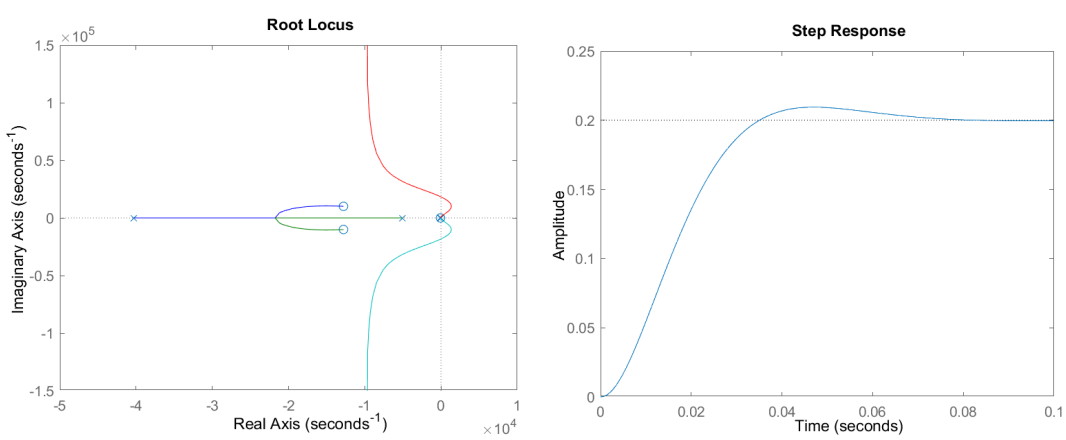

Figure 3.5: a) Root-locus and b) Step response of the deflection states

- Four of the six poles clustered around -65 with the other two 4 orders of magnitude away
- The step response is found to be damped for all wind speeds.

b. Combined System

- All poles and zeros appear on the left half plane. However, one of the flap angle poles is located much closer to the imaginary axis and dominates the overall system dynamics.

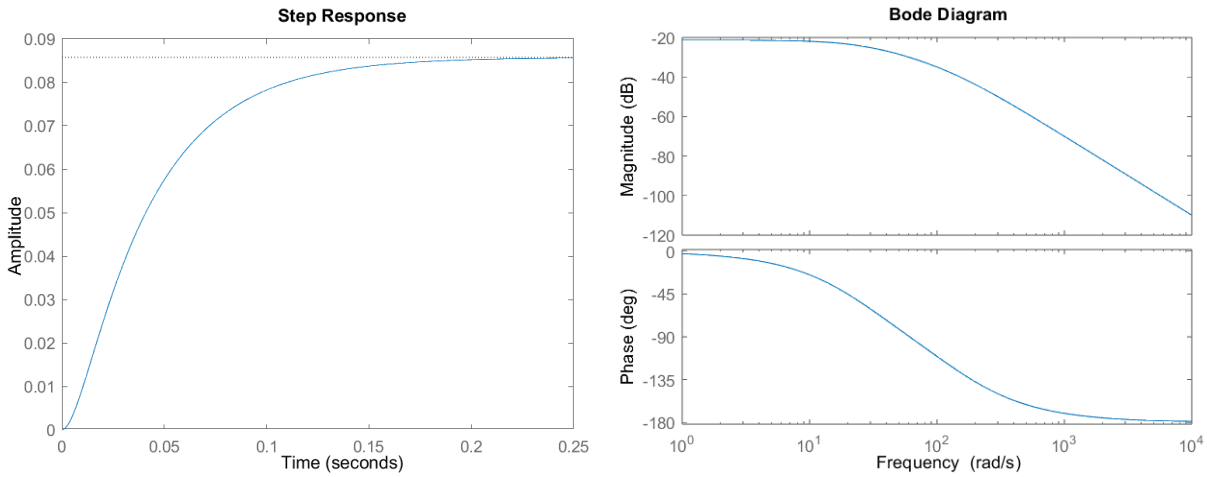

Figure 3.6: a) Step response and b) Bode-plot of the combined state

- Bode Plot suggests the system does not produce any significant response to input frequencies above 100 rad/s and always lags the input

CHAPTER 4 - Control Design

4.1 Control Architecture

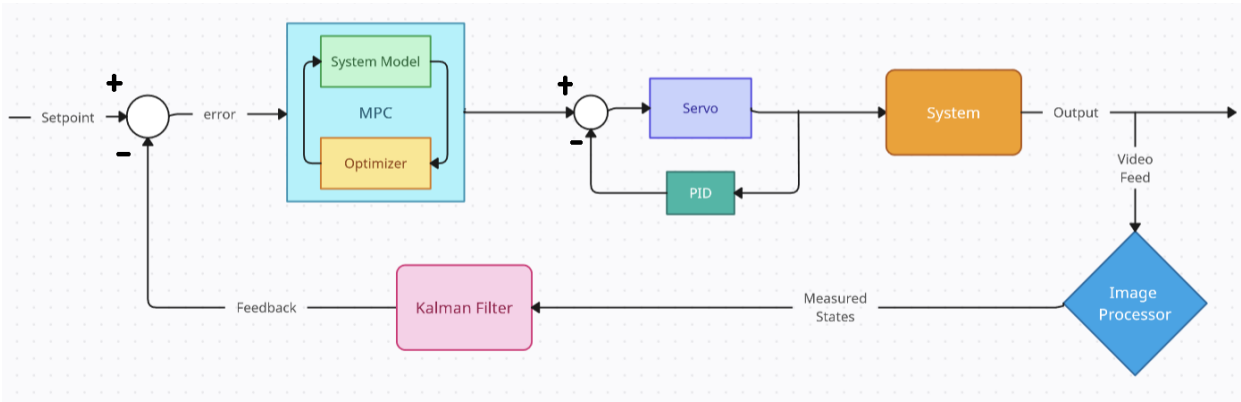

Figure 4.1: Control system architecture

The setpoint error is used to calculate the optimal control move in the form of commanded morphing angle (β_c) or shaft angle. This is sen to the servo motor, which is further controlled by an inner loop PID. The shaft angle is transformed into the morphing angle (β) by the system. The morphed position is recorded by a camera and processed in the image processor to extract measurements. These are then sent to the Kalman filter for estimation of unmeasured states. The estimated states are feedback into the system in the form of a setpoint error, closing the loop.

4.2 Model Predictive Control

A constrained MPC was used to determine the optimal control move. The model was discretized with a sampling time of 0.33s to match the incoming video feed. Constraints are applied on the commanded flap angle to limit actuation between -11° and 20°. Np and Nc were tuned to minimize control effort while reaching the setpoint in the lowest possible time. A shared system model is used for the MPC and Kalman filter.

The overall execution follows the following sequence:

a. Obtain measurements if available

b. Obtain current states using the Kalman Filter

c. Compute optimal control moves

d. Implement the first move

Repeat

4.3 Measurement and Estimation

Position of the free end of the morphing section is needed by the control algorithm. However, it is difficult to measure by conventional means as the tip deflects in both transversal and axial directions. A video-based measurement technique is developed.

The Centerline of the morphing section is marked using red patches. A camera records and send the data to an image processor, which uses a color-based clustering algorithm to find the centroids of the red patches. A spline fit is used to recover the complete deflection profile.

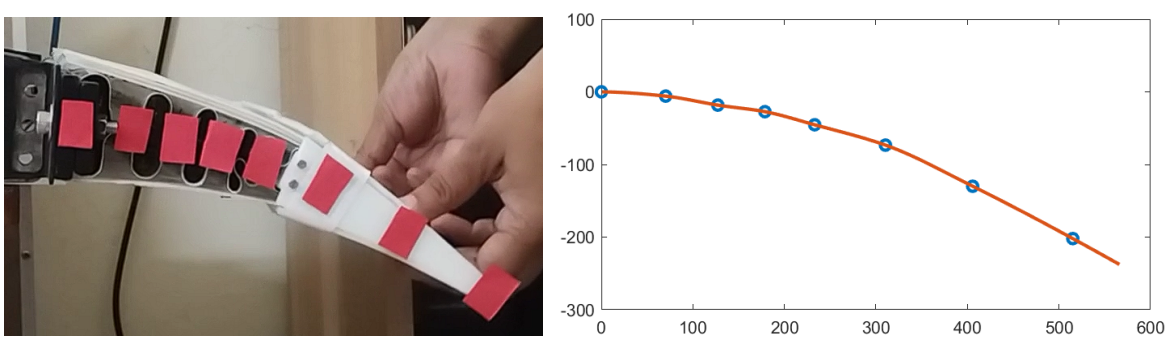

Figure 4.2: a) Deflection of the morphing section b) Post-processed deflection of the centerline

The deflection states are obtained using a regression fit using the DCVC mode shapes. A Kalman filter is then used to estimate the remaining states.

The filtering algorithm can be split into a predictor and corrector step, details of which are shown in figure 12. Here, P_k is the error covariance matrix and K_k is the kalman gain at k^{th} timestep.

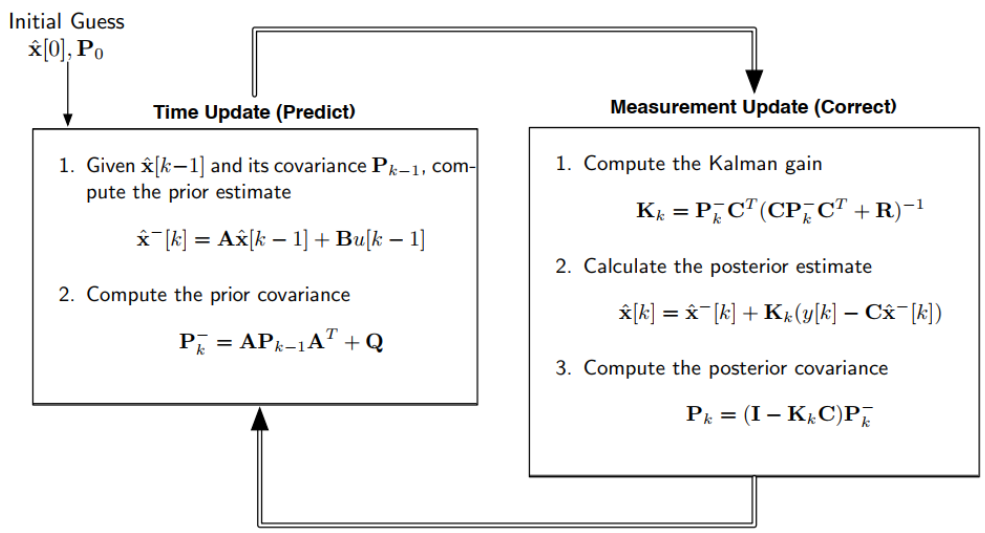

Figure 4.3: Prediction Correction algorithm in a Kalman Filter state estimator [7]

4.4 Results and Discussion

Note: All simulations were done for U = 15 m/s and α = 5° with a sampling time of 0.033s

a. To appreciate the benefits of MPC over classical control techniques, we simulate an MPC with (Np, Nc) = (1, 1), which can be considered to represent classical control methods like PID.

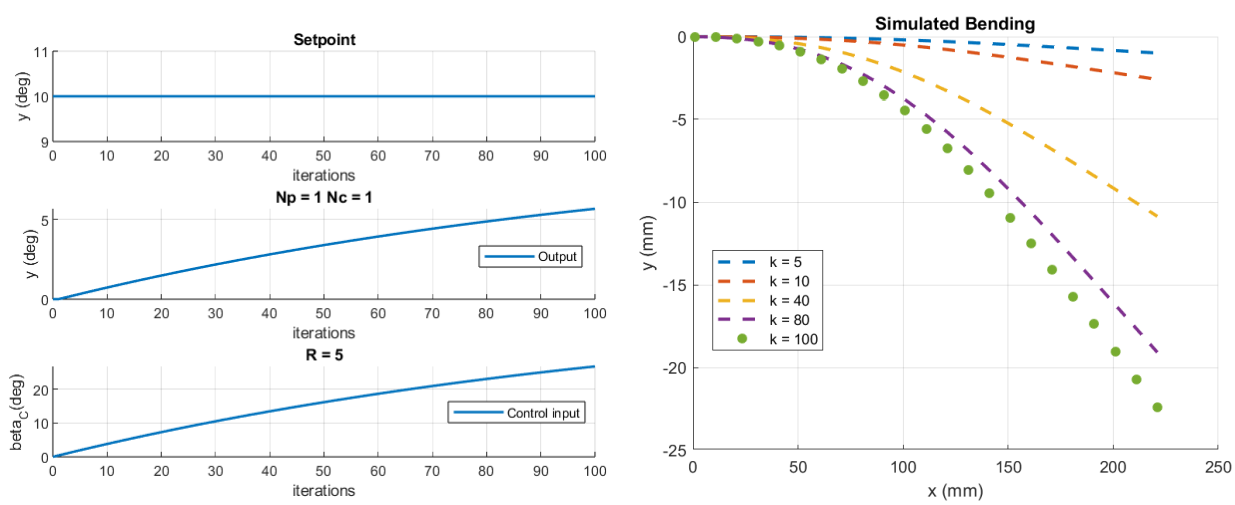

Figure 4.4: a) Setpoint, output and control move for (Np, Nc) = (1, 1)

b) Corresponding simulated deflection at different time steps

We observe the sluggish response from the system. Further, the control input has exceeded the input limit. A PID controller cannot internally accommodate physical constraints.

b. Using a prediction horizon (Np) of 5 timesteps, control horizon (Nc) of 2 timesteps, and control cost (R) of 0.5 is used. The setpoint is achieved within five timesteps (0.17s) and the limit on control input is also not hit. The simulated bending is in close agreement with the setpoint.

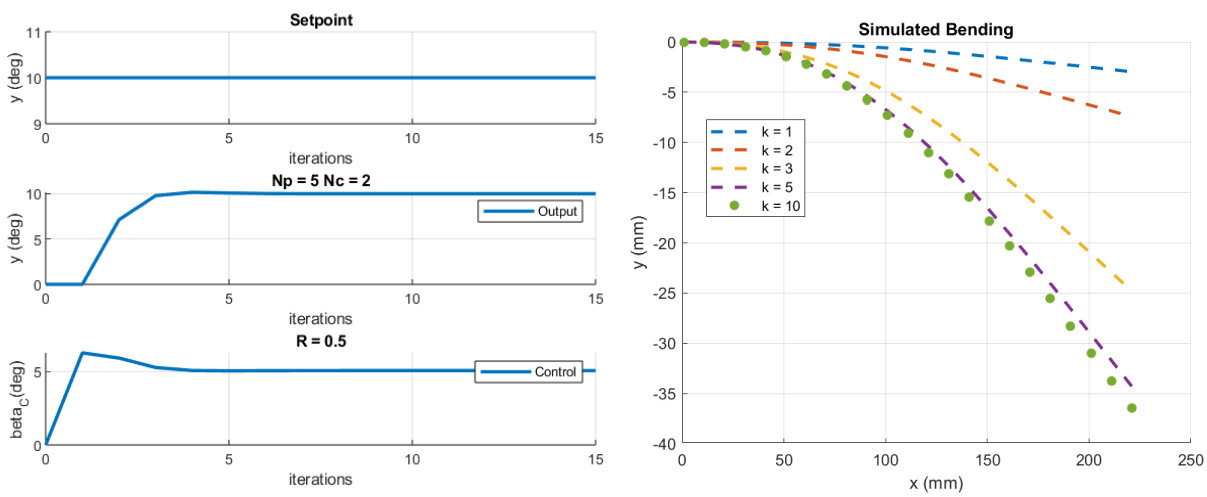

Figure 4.5: a) Setpoint, output and control move for (Np, Nc) = (5, 2)

b) Corresponding simulated deflection at different time steps

c. To observe the working of the Kalman filter, a noise component is added to the simulated output and fed into the system as measurement. We define SNR as the ratio of the amplitude of output to noise. (As the setpoint is deterministic, a variance-based definition cannot be used). Further we define control cost to be the area under the incremental control move (△u).

Q and R are the tuning parameters for the Kalman filter and represent the designers distrust in the model and measurement respectively.

Note: Q and R for the Kalman Filter are mentioned above the output plot. R (penalty associated with control effort) is mentioned above the control input plot.

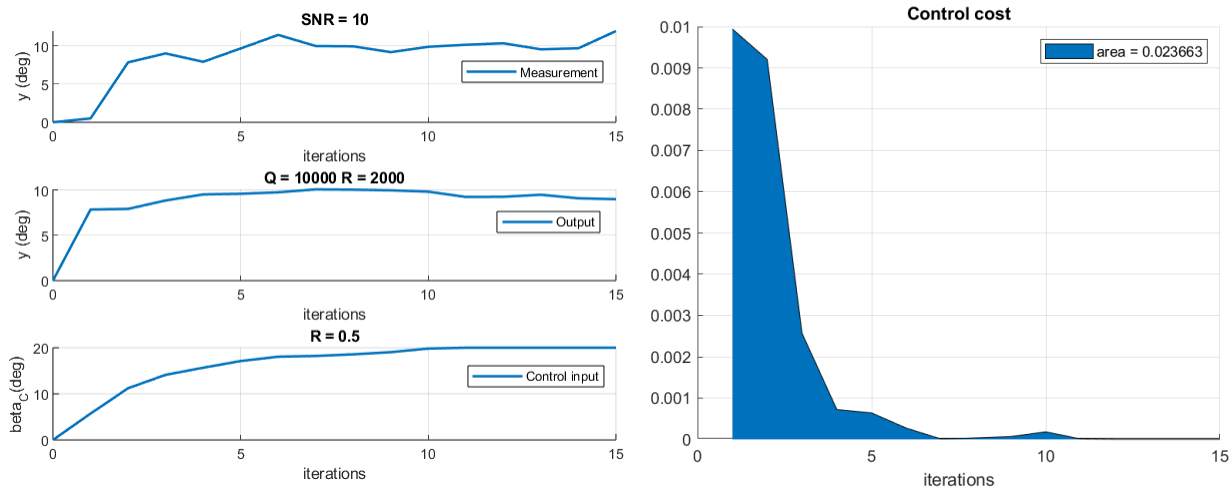

Figure 4.6: a) Setpoint, output and control move for SNR = 10

b) Corresponding control cost associated

For an SNR of 10, the Kalman filter fares well to estimate the true state. However, the control cost is observed to be nearly two times the noise-free case.

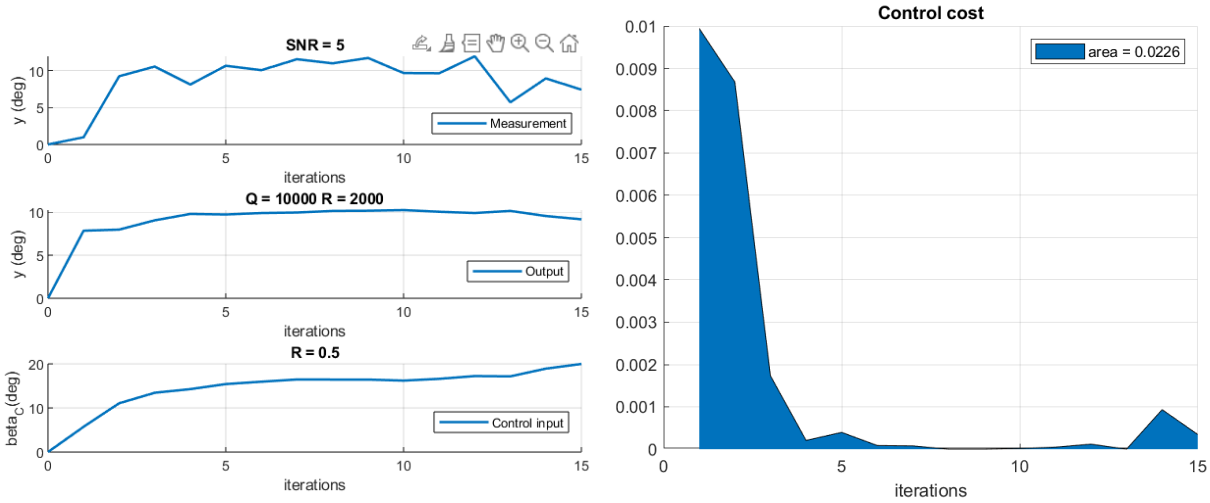

Figure 4.7: a) Setpoint, output and control move for SNR = 5

b) Corresponding control cost associated

For an SNR of 5, the Kalman filter still estimates the state correctly without much-added control cost. However, reducing the further introduces a bias in the output. The magnitude of this bias can be changed by adjusting Q and R

CHAPTER 5 - Conclusion and Future Work

5.1 Significance of current work

With a pant model and controller in place, the system is better informed of the aerodynamic loads, morphing section dynamics, and its structural limitations. The controller can be tuned to achieve performance (faster response) or energy economy, and the battery and power supply can be sized accordingly.

The morphing section deflects in two dimensions, making it fundamentally difficult to measure using conventional sensors like LVDTs and optical sensors. The video-based measurement and post-processing technique solves this problem and extracts the complete deflection profile, allowing for full-state feedback.

5.2 Problems Identified

A number of assumptions were made to develop the system model, but these might not necessarily be true.

- Euler Bernoulli equation for bending assumes small deflection, and the analytical solution is valid only for homogenous beams.
- For the calculation of aerodynamic loads, the pressure distribution needs to be integrated over the camber line. However, for simplicity, the camber line of the morphing section was assumed to coincide with the chord.
- Several parameters specific to the setup and required for the simulation were not readily available. Hence sample values for literature were used.

5.3 Scope for future study

- A large deflection model that accounts for the cord-wise variations of stiffness and inertia can be developed for the system model. (The iterative model developed by Deepak et al. (2021) [3] cannot be used for this as it is too computationally expensive for online estimation)
- Aerodynamic loads need to be re-evaluated for the unsymmetric airfoil
- The system parameters need to be estimated
- The control system has not been physically tested with the lab setup. The MPC and servo PID should be tuned before wind tunnel (load) testing.

List of References

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[2] B. Marco and Toropov. “Aeroelastic Analysis and Gust Response of a Flexible Airfoil.” Collection of Technical Papers - AIAA/ASME/ASCE/AHS/ASC Structures, Structural Dynamics and Materials Conference, Orlando, Florida, 15 Apr 2010.

[3] D. Kumar, S. F. Ali, A. Arockiarajan. “Theoretical and experimental studies on large deflection analysis of double corrugated cantilever structures.” International Journal of Solids and Structures, (2021), Volume 228, Article 111126.

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[5] D. Hodges, and G. Pierce. Introduction to Structural Dynamics and Aeroelasticity (2nd ed., Cambridge University Press, Aerospace Series, New York, USA, 2011)

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[7] L. Wang. Model Predictive Control System Design and Implementation using MATLAB (Advances in Industrial Control, Springer, Melbourne, Australia, 2009)

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Appendix A

1 Structural Dynamics

a) Transverse vibrations of a beam:

Governing Equation:

$$\boxed{\frac{\partial^2}{\partial x^2}\left(EI\frac{\partial^2 w}{\partial x^2}\right) = -\mu\frac{\partial^2 w}{\partial t^2} + q(x).}$$

Under assumptions of constant E and I, an analytical solution can be obtained for the deformation(w) using separation of variables:

$$w(x,t) = \sum_{i=1}^{\infty}\phi_i(x)\xi_i(t)$$

For a cantilever beam, the boundary conditions are as follows:

$$w(0,t) = 0 \qquad w'(0,t) = 0 \qquad w''(l,t) = 0 \qquad w'''(l,t) = 0$$

The mode shapes thus obtained are:

$$\hat{w}_n = A_1\left[(\cos\beta_n x + \cosh\beta_n x) - \frac{\cos\beta_n L - \cosh\beta_n L}{\sin\beta_n L - \sinh\beta_n L}(\sin\beta_n x + \sinh\beta_n x)\right]$$

Where β_n is found by solving the characteristic equation:

$$\cosh(\beta_n L)\ \cos(\beta_n L) + 1 = 0$$

b) Aerodynamic load on the morphing section:

$$L_{morph} = \int_0^l P_{nc}(x,t)\cos\theta\, dx = \int_0^l \frac{\rho_f U_\infty^2(2x-l)}{\sqrt{x(l-x)}}\frac{\partial w}{\partial x}\cos\left(\frac{\partial w}{\partial x}\right)dx$$

For small deflection ($\theta<20^o$) we can approximate:

$$\cos\theta = \cos\left(\frac{\partial w}{\partial x}\right) \approx 1$$

$$L_{morph} = \int_0^l \frac{\rho_f U_\infty^2 (2x - l)}{\sqrt{x(l-x)}} \frac{\partial w}{\partial x} dx = \frac{\rho_f U_\infty^2}{EI} \int_0^l \frac{(2x - l)}{\sqrt{x(l-x)}} x \, dx M_{in}$$

$$M_{in} = K_\beta . \beta_c$$

The ith modal force Q_i modal force is found by multiplying dL_{morph} with the ith mode shape ($\Phi_i(x)$) and integrating over the length of the morphing section

c) State Matrices for the combined model

- Phase transition matrix

$$A = \begin{bmatrix} O_{3x3} & I_{3x3} & O_{3x1} & O_{3x1} \\ -M_m^{-1} \cdot K_m & -M_m^{-1} \cdot K_m & O_{3x1} & O_{3x1} \\ O_{1x3} & O_{1x3} & 0 & 1 \\ O_{1x3} & O_{1x3} & -K_\beta / I_\beta & -C_\beta / I_\beta \end{bmatrix}_{8x8}$$

Here, I_m, C_m, and K_m, represent modal mass, damping, and stiffness and are 3x3 matrices

- Control Matrix

$$B = \begin{bmatrix} O_{3x1} \\ M^{-1} Q \\ 0 \\ K_\beta / I_\beta \end{bmatrix} \qquad Q = \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \end{bmatrix}$$

- Measurement Matrix

$$C = [O_{6x1} \quad 1 \quad 0]$$

- There is no feedthrough term, hence D = 0.

Note: O_{axb} and I_{axb} represent the zero matrix and identity matrix respectively of dimensions a x b

Appendix B

1. Link to Project folder:

   https://drive.google.com/drive/folders/1b6Bo1RAnfhFRyQlveoKNpD9IvbV1bqMH?usp=share_link

2. Link to Poster presentation:

   https://drive.google.com/file/d/1iGsoNfJF0YBozJjncldyemD3xvLH76Po/view?usp=share_link